

Year 7 • Lesson 10.02 Misleading graphs: answers (teacher copy)

10.02 Misleading graphs (booklet p. 3)

Warm-up

1. 6
2. 3
3. 4
4. A title

WE DO examples

1. Ex 1: the vertical axis starts at 46, not 0, so Ava's lead looks huge
2. Ex 2: $52 - 48 = 4$ more votes (not 3 times as many)
3. Ex 3: Graph P, because its axis starts at 115 so the bars' heights show only the part above 115
4. Ex 4: No. 130 is only 10 more than 120
5. Ex 5: column graph starting at 0 with even scale, title and labels
6. Ex 6: Ava or her supporters, to make her win look bigger than it is

Set A

1. F
2. M
3. M
4. F
5. M
6. F
7. M
8. M
9. F
10. M

Set B

1. \$41 000
2. 2020
3. 39
4. Yes
5. \$6000 (6 thousand)
6. About 7 times
7. False
8. 2020 to 2021 (+3)
9. Yes
10. The vertical axis starts at 39, not 0

Set C

1. 115
2. 0
3. 10
4. 3 times
5. Graph P, because the increase looks much bigger
6. 110%
7. The percentages add to more than 100%
8. 10 years
9. 2 years
10. Equal gaps show different lengths of time, so the growth looks like it slows down (it is really steady)

Set D

1. Brand C looks about 3 times Brand B
2. B 200 000, C 210 000; difference 10 000
3. Fair column graph from 0
4. Fair line graph from 0
5. The rise looks small and steady, not dramatic
6. No: 54 is not double 52. The rise is only 2
7. It looks 4 times bigger, not 2 times
8. e.g. to sell a product, win a vote, make results look better or worse
9. e.g. title, labelled axes with units, scale starting at 0 with even gaps
10. The missing months could show sales going down, so the trend looks better than it is

Extension

1. 9 times
2. $1.5 \times 1.5 = 2.25$ times bigger; draw it the same size and use $\frac{1}{2}$ pictures, or make only one direction bigger

Exit ticket

1. Level 1: 80
2. Level 2: 4 points
3. Level 3: the axis starts at 80, so the bars exaggerate the difference; start the axis at 0